THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 03

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 4 units	distance from the x-axis is $\lvert y\rvert$
2	**(a)** $(4,\,-4)$	Quadrant II needs $x<0,\ y>0$
3	**(d)** Quadrant III	signs of (x,y) are $(-,-)$
4	**(b)** 0	highest power of the variable is 0
5	**(b)** parallel	equal slopes, different intercepts
6	**(b)** irrational	12 is not a perfect square
7	**(a)** $\mathbb{Q}$	Q for quotient
8	**(a)** -5	Brahmagupta's rules for signs
9	**(b)** 6	$x^2, y^2, z^2, 2xy, 2yz, 2zx$
10	**(b)** $25x^2-10x+1$	$(a-b)^2=a^2-2ab+b^2$
11	**(d)** 16 cm	half-chord $=\sqrt{10^2-6^2}=8$, so chord $=2\times 8=16$
12	**(a)** 25 cm	$r=\sqrt{24^2+7^2}=25$
13	**(d)** 35°	angle at the centre is double the angle on the circle
14	**(a)** 154 cm^2	$\pi r^2=\frac{22}{7}\times 7^2=154$
15	**(d)** 33 cm	$\frac{90}{360}\times 2\pi r=33$
16	**(a)** 120 cm^2	area $=bh$
17	**(a)** $\frac{1}{3}$	favourable outcomes: 3, 5 ⇒ 2 out of 6
18	**(b)** $\frac{3}{10}$	6 favourable out of 20 equally likely balls
19	**(d)** Prakrit metre and poetry	7th century CE, Vṛttajātisamuchaya
20	**(c)** 1275	$\frac{50\cdot 51}{2}$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$\frac{-4+x}{2}=2\Rightarrow x=8$; $\frac{3+y}{2}=-1\Rightarrow y=-5$. So $B=(8,\,-5)$	1+1
22	$2(7)+1=15$ ✓. $2(3)+1=7\neq 8$ ×.	1+1
23	$4\times 15=60$ and $6\times 10=60$, so $ad=bc$ ⇒ equal. (Both equal $\frac{2}{3}$.)	1+1
24	$(x-4)\,(x-3)$	2
25	Chord at 3 cm: $2\sqrt{25-9}=8$ cm; chord at 4 cm: $2\sqrt{25-16}=6$ cm. The chord nearer the centre is longer.	1+1
26	$s=6$; area $=\sqrt{6\times 3\times 2\times 1}=6$ cm^2. Right-angled: $\frac{1}{2}\times 3\times 4=6$ ✓	1+1
27	$\frac{112}{200}=0.56$. The theoretical value is 0.5; with more tosses the experimental value tends to get closer.	1+1
28	Ratios $\frac{6}{2}=\frac{18}{6}=\frac{54}{18}=3$, a constant (1). $t_n=2\cdot 3^{n-1}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Let $P(0,y)$. $PA^2=36+(y-5)^2$, $PB^2=16+(y-3)^2$. $36+y^2-10y+25=16+y^2-6y+9\Rightarrow 61-10y=25-6y\Rightarrow y=9$. $P=(0,9)$	1+1+1
30	$500-20d$; zero when $d=25$ days. After 15 days, 200 pages remain.	2+1
31	$15\div 2=\frac{15}{2}=7\frac{1}{2}$: 7 kurtas, with 1 m left over.	1+1+1
32	Area $=(2x+3)^2=4x^2+12x+9$ sq. units; perimeter $=4(2x+3)=8x+12$ units.	2+1
33	30 min = half a turn: arc $=\pi r=44$ cm (1). Area $=\frac{1}{2}\pi r^2=\frac{1}{2}\times\frac{22}{7}\times 196=308$ cm^2 (2).	1+2
34	Only 2 is both even and prime ⇒ $\frac{1}{6}$ (1). The numbers that are even or prime are 2, 3, 4, 5, 6, so only 1 is neither ⇒ $\frac{1}{6}$ (2).	1+2
35	Half-chord $=\sqrt{r^2-d^2}$ (Baudhāyana–Pythagoras) (1); chord $=2\sqrt{r^2-d^2}$ (1). For $r=10, d=6$: $2\sqrt{64}=16$ (1).	1+1+1

	OR: In a parallelogram opposite angles are equal: $\angle A = \angle C$ (1). In a cyclic quadrilateral $\angle A + \angle C = 180^\circ$ (1). So each is $90^\circ \Rightarrow$ rectangle (1).	1+1+1
36	$1, 5, 13, 29, 61, 125, 253, \ldots$ (2). 133 does not appear and the terms increase, so it is not a term (1).	2+1
	OR: $18,\ 13.5,\ 10.125$ ft (2). It is a GP with $r = \frac{3}{4}$ (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) -12 and 12. (ii) $-850 + 1200 - 450 = -100 \Rightarrow$ debt of ₹100. (iii) Removing 4 debts of ₹3 each makes one ₹12 richer.	2+2+1
	OR: (i) For $a < b$ the average $\frac{a+b}{2}$ is rational and lies between them (3). (ii) $\frac{1}{2}\left(-\frac{3}{12} + \frac{5}{12}\right) = \frac{1}{12}$; it lies to the right of 0, one-twelfth of a unit from the origin (2).	3+2
38	(i) 10 m rope: $\sqrt{169 - 25} = 12$ m; 24 m rope: $\sqrt{169 - 144} = 5$ m (2). (ii) $12 + 5 = 17$ m (1). (iii) $12 - 5 = 7$ m (1). (iv) 24 m rope is longer and nearer to the centre (1).	2+1+1+1
	OR: $CA = CD \Rightarrow \angle ACE = 2\angle ADC$ (exterior angle) (2). $CB = CD \Rightarrow \angle BCE = 2\angle BDC$ (2). Subtract: $\angle ACB = \angle ACE - \angle BCE = 2(\angle ADC - \angle BDC) = 2\angle ADB$ (1).	2+2+1
39	(i) Construct a square with the same area as the rectangle (1). (ii) Area $= 100$ cm^2; side $= 10$ cm (2). (iii) Square perimeter $= 40$ cm; rectangle perimeter $= 58$ cm — the square has the smaller perimeter for the same area (2).	1+2+2
	OR: (i) $s = 88$ m (1). (ii) $\sqrt{63 \times 49 \times 36 \times 28} = 1764$ m^2 (2). (iii) $25^2 + 60^2 = 65^2$ and $39^2 + 52^2 = 65^2$, so both triangles are right-angled; areas $750 + 1014 = 1764$ ✓ (1). (iv) $1764 \times 20 =$ ₹35,280 (1).	1+2+1+1
40	(i) 100, 150, 200 (1). (ii) $t_n = 100 + 50(n-1) = 50n + 50$ (1). (iii) $t_{12} = 650$; total $= \frac{12}{2}(100 + 650) =$ ₹4,500 (2). (iv) $50n + 50 = 1000 \Rightarrow n = 19$ (1).	1+1+2+1
	OR: (i) 1000, 2000, 4000, 8000 (1). (ii) $500 \cdot 2^n$ (1). (iii) $500 \times 64 = 32000$ (1). (iv) $500 \cdot 2^n > 10000 \Rightarrow 2^n > 20 \Rightarrow n = 5$ (16,000) (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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